Neural Networks - Task 2

Penguin classification using backpropagation

1. Introduction

The aim of this task was to classify three penguin species using a multilayer neural network trained with backpropagation. Sigmoid and tanh were compared using the same dataset. The network uses five input features and three output neurons, with one output for each species.

2. Experiment setup

A total of 48 configurations were tried, using 1, 2, 3, 4, 5 hidden layers. The learning rates were 0.01, 0.05, and the epoch counts were 600, 1000. Weights and enabled biases were initialized with small random values between -0.1 and 0.1.

To compare the settings, 24 of the 30 training samples per species were used to fit each network and the other 6 were used for validation. Preprocessing was fitted on those 72 training samples. The configuration with the highest validation accuracy was selected for each activation; lower validation loss was used to break ties. Each selected network was then retrained on all 90 training samples before testing on the remaining 60.

3. Results

Activation	Train accuracy	Test accuracy	LR	Epochs	Hidden layers	Neurons per layer	Bias	Seed
Sigmoid	98.89%	98.33%	0.01	600	1	12	On	42
Tanh	100.00%	96.67%	0.05	1000	2	12, 8	On	42

The sigmoid network classified 59 out of 60 test samples correctly (98.33%). The tanh network classified 58 out of 60 test samples correctly (96.67%).

Sigmoid results

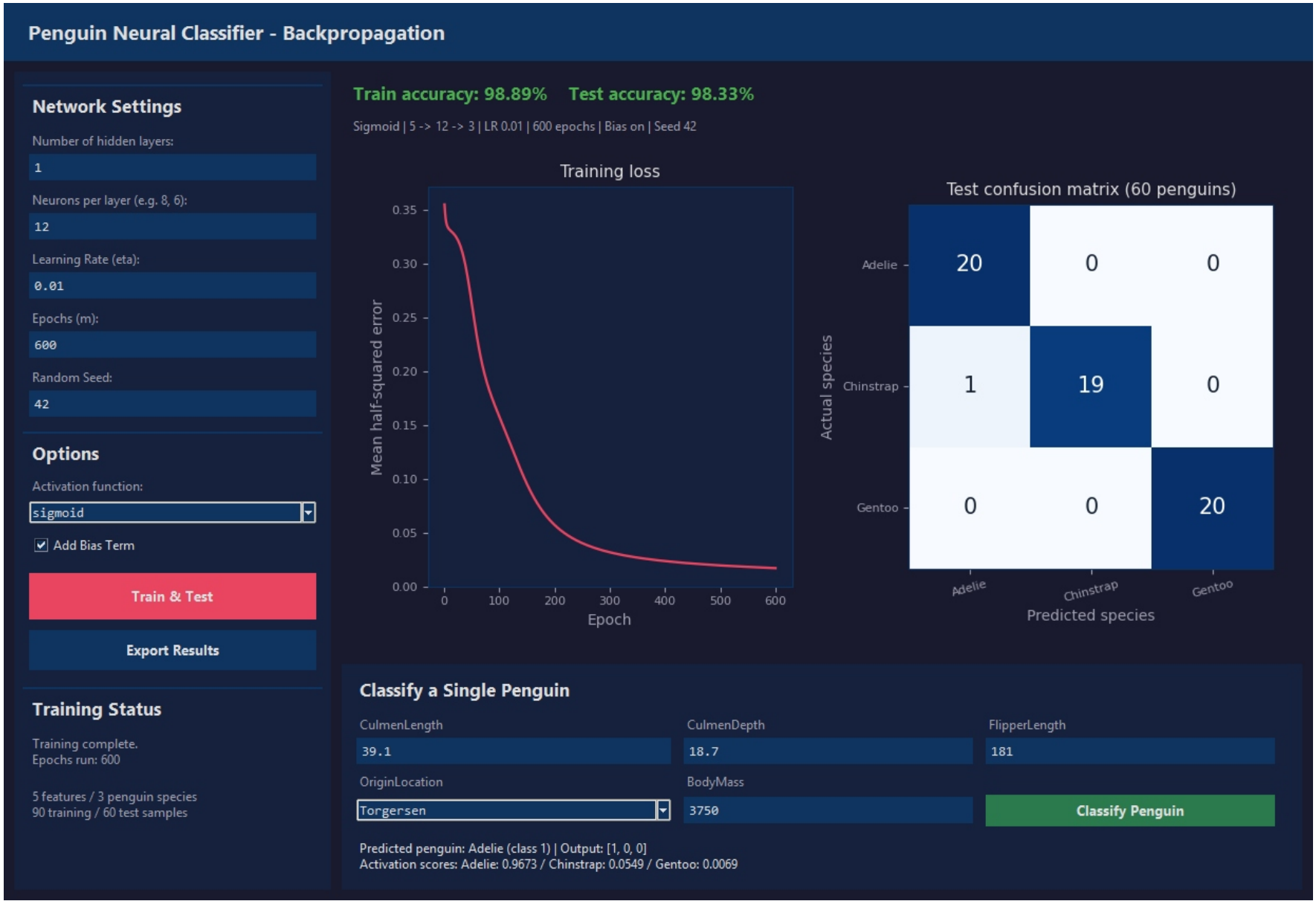


Figure 1. Settings, training loss and test confusion matrix for the sigmoid network.

Tanh results

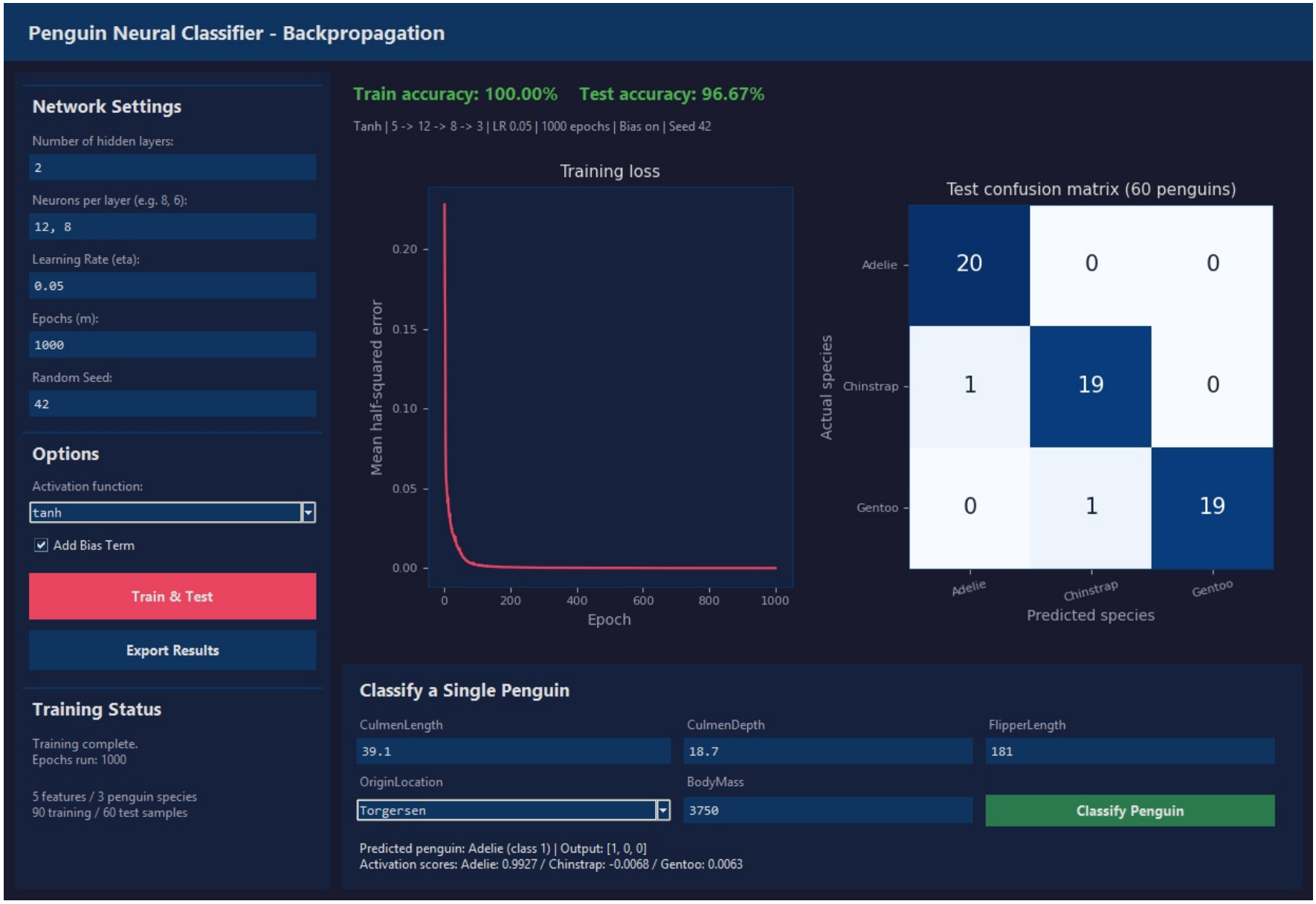


Figure 2. Settings, training loss and test confusion matrix for the tanh network.